Data Science Demystified: Advanced Concepts Made Simple

Requirements and Technical Setup

01

Optional Column Top Line

Visual Studio Code installation

VS code is an open-source code editor which is lightweight, fast, and supports a wide range of programming languages through extensions

02

Optional Column Top Line

Python installation and setup

Python is a popular, high-level programming language known for its readability, simplicity, and versatility.

03

Optional Column Top Line

Virtual Environment setup

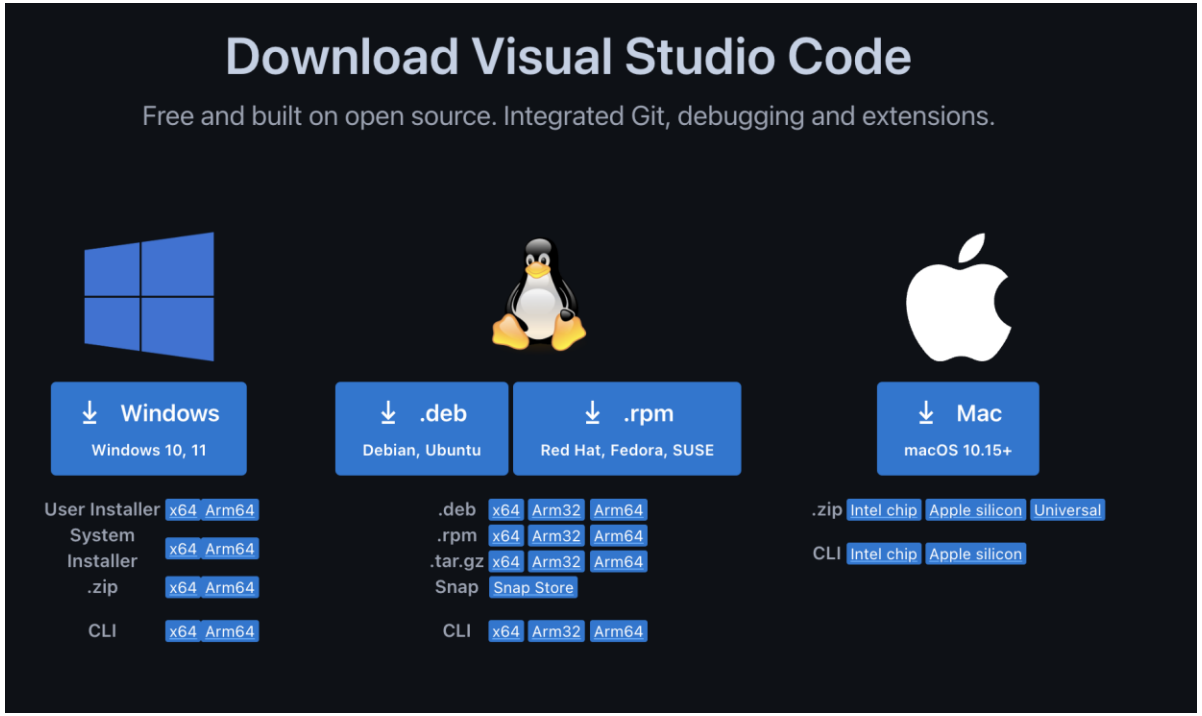
Each virtual environment has its own Python interpreter and libraries and helps to create isolated virtual environments for individual projects

Visual Studio Code installation and Setup



Visual Studio Code installation instructions

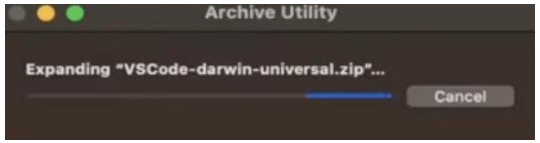
1. Visit the official VS Code page at https://code.visualstudio.com/
2. Download VS Code for Mac or Windows based on your machine's operating system

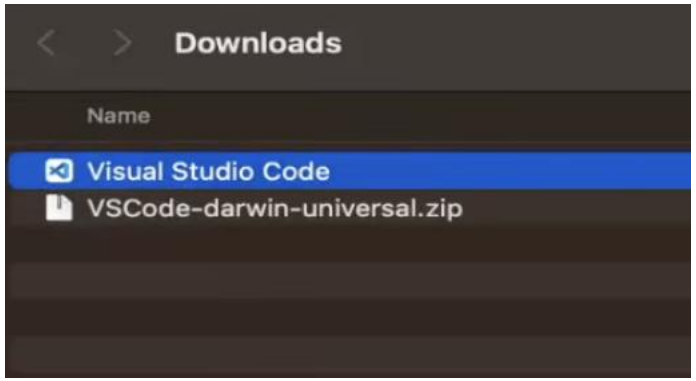


VS Code step-by-step guide for Mac

Double click the downloaded **.zip** file from the ‘Downloads’ folder

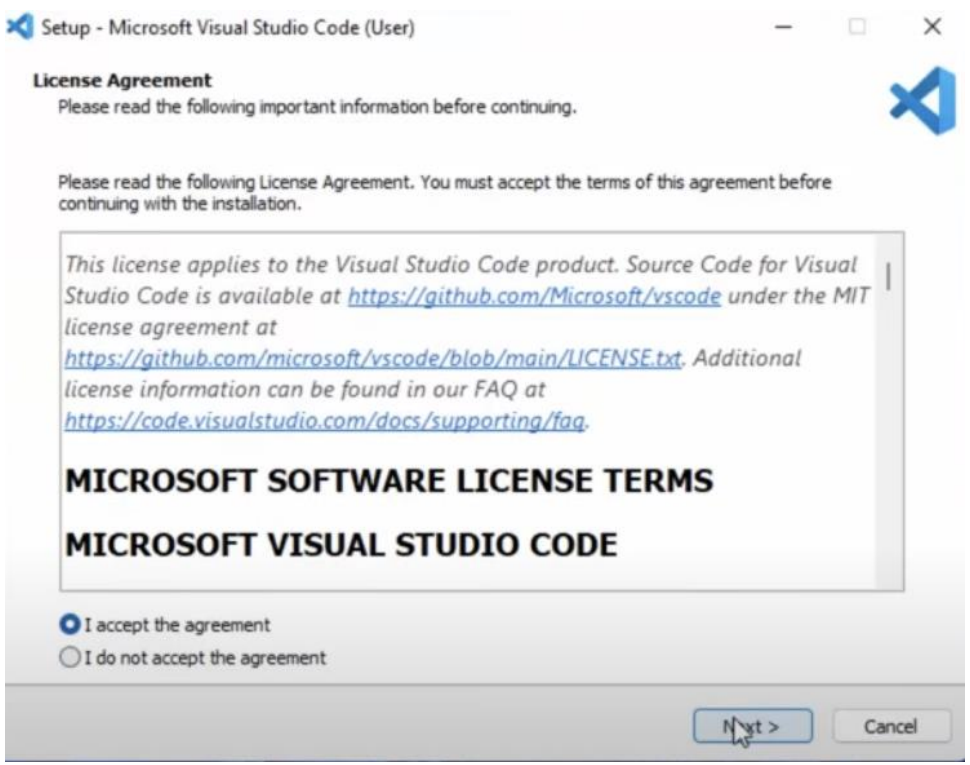


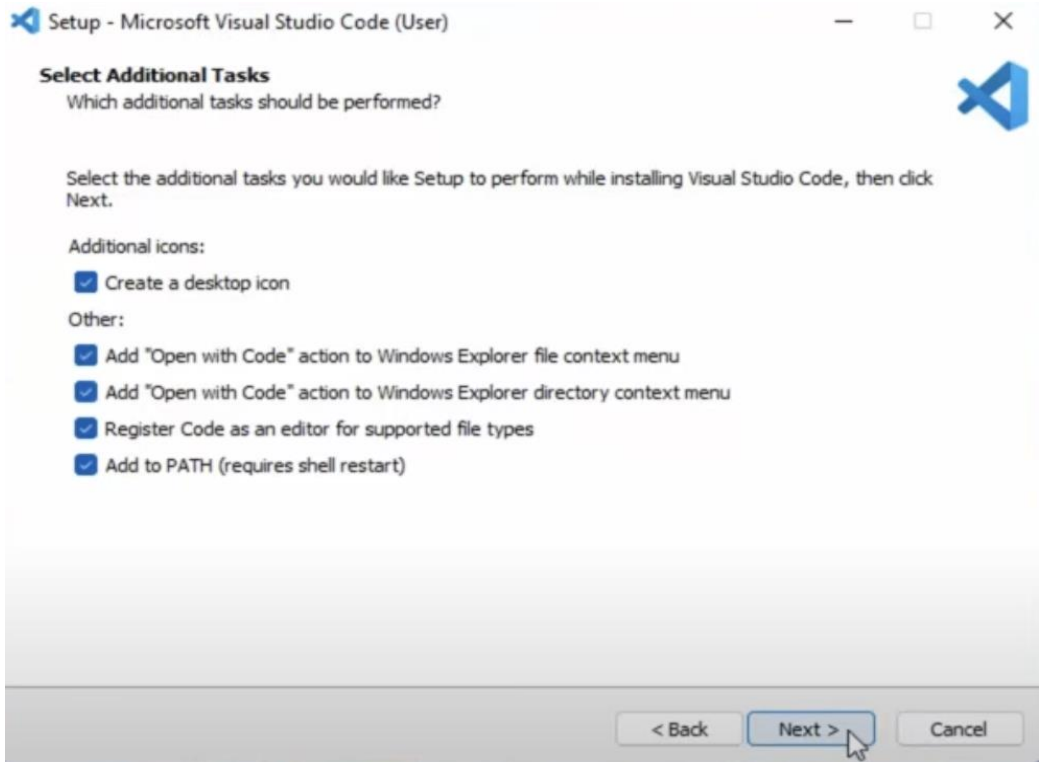




VS Code step-by-step guide for Windows

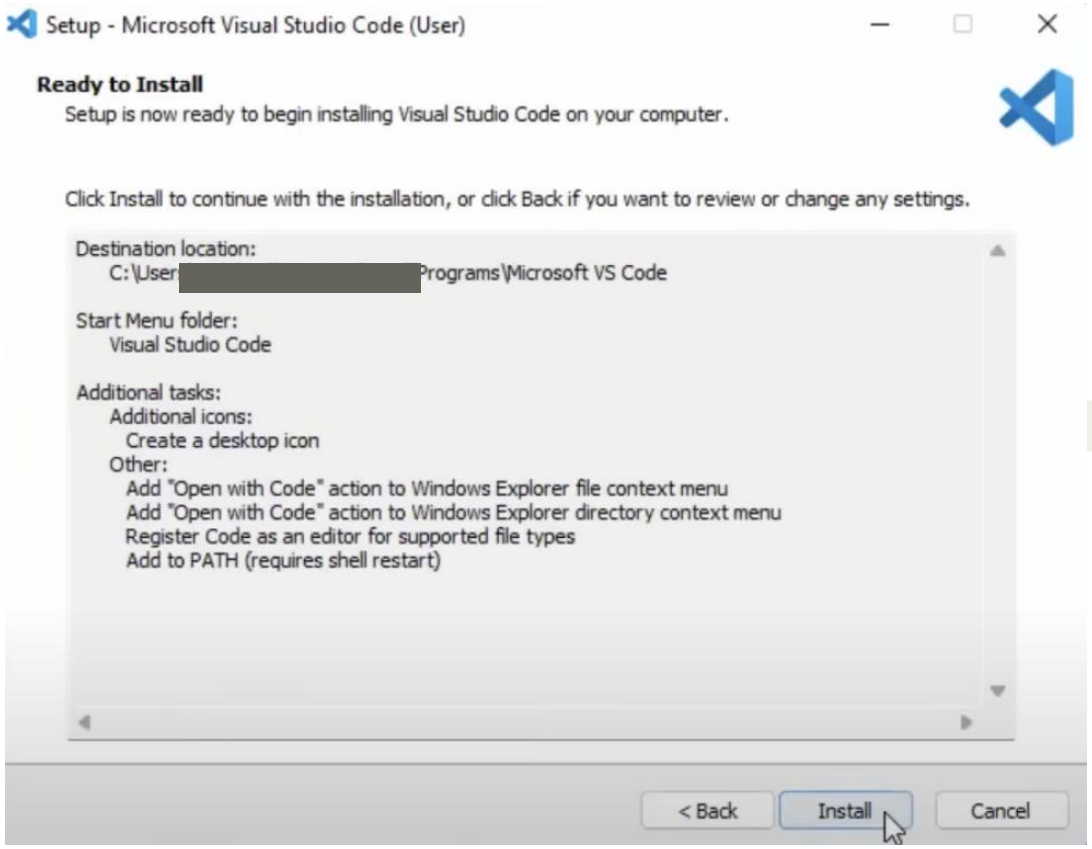
1. Double click the downloaded **.exe** file from the ‘Downloads’ folder
2. **Do not change** the default values for **Destination folder and Start Menu Folder**, if prompted
3. Check all additional tasks

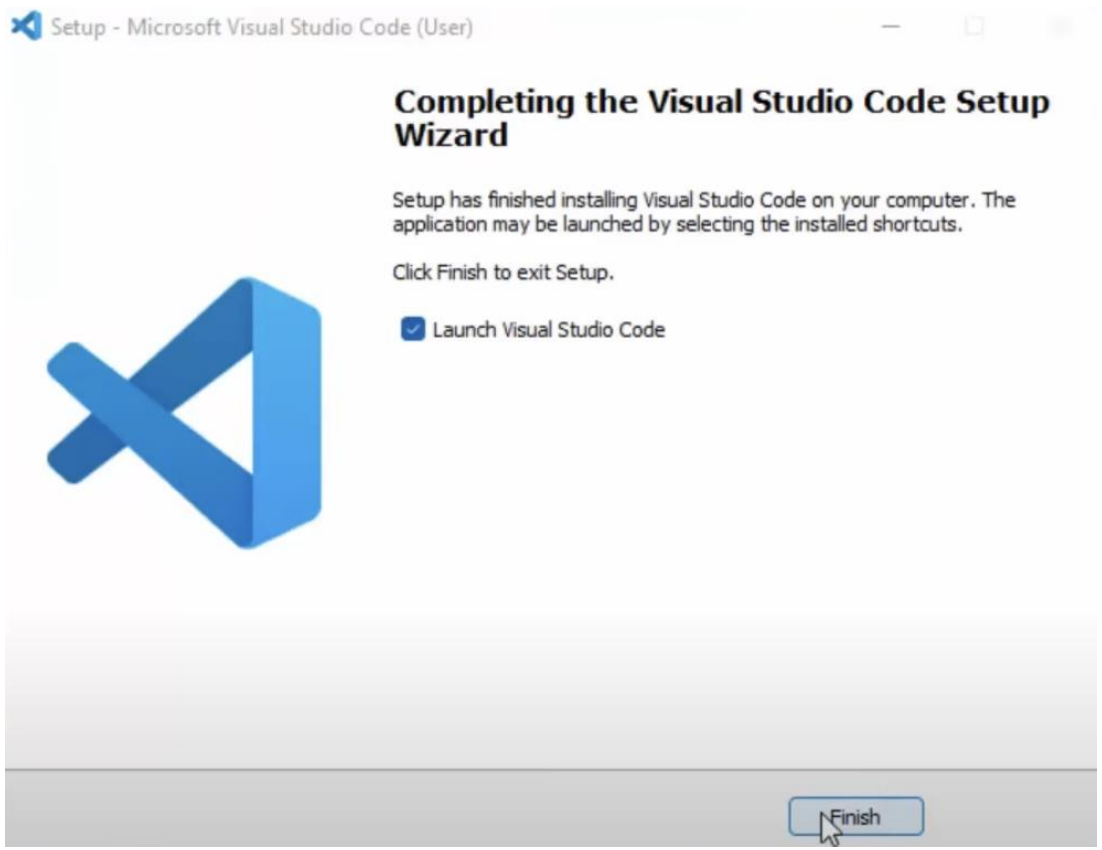




VS Code step-by-step guide for Windows

4. Click '**Install**' and '**Finish**'



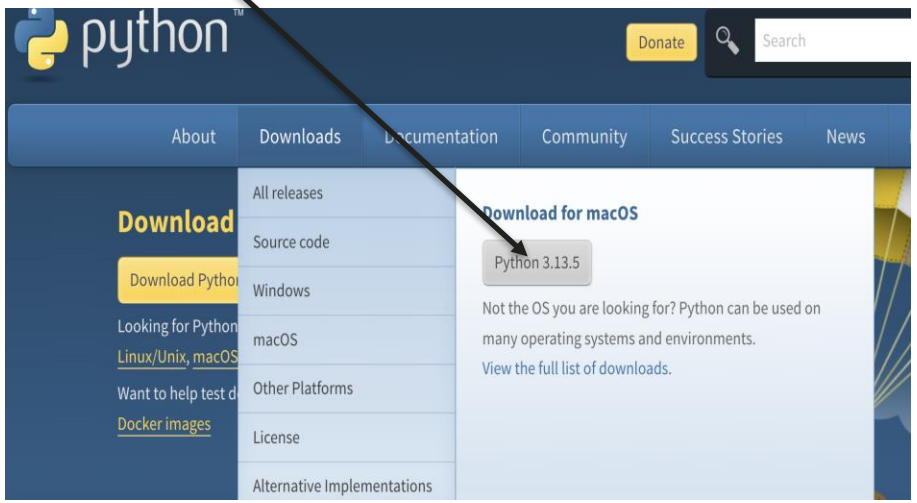


Python installation and setup



Python installation step-by-step guide

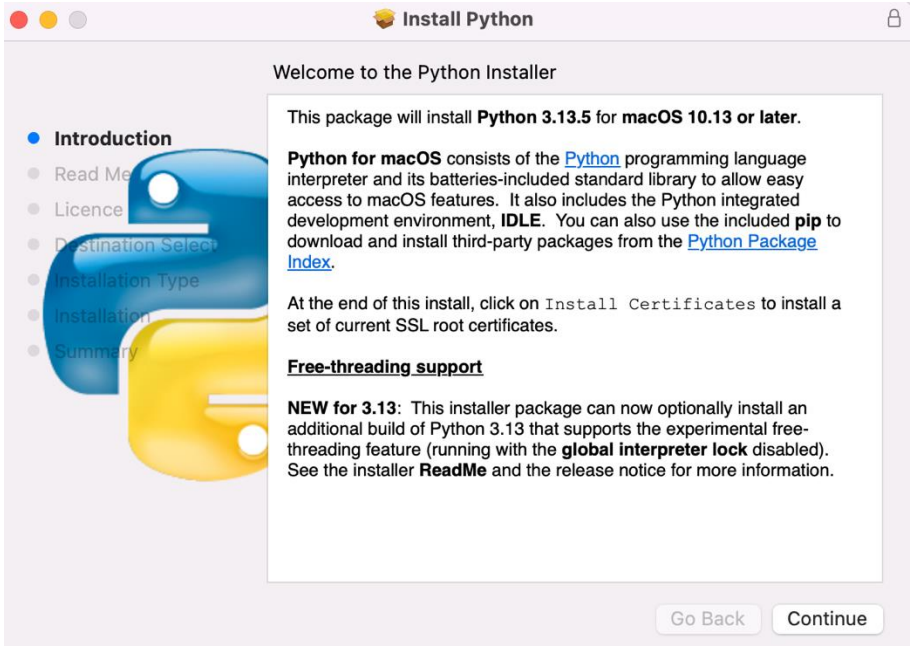
1. Visit python's official downloads page at https://www.python.org/downloads/
2. Place the mouse cursor on the 'Downloads' menu
3. Click **here** to download python 3.13.5 for Mac or Windows based on your machine's operating system

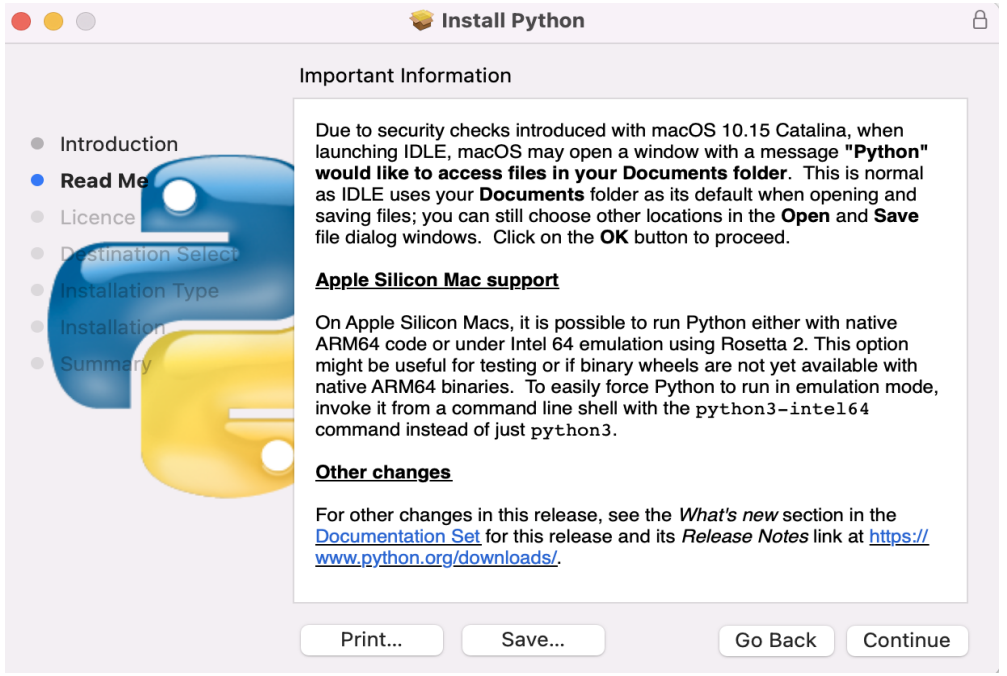




Python step-by-step guide for Mac

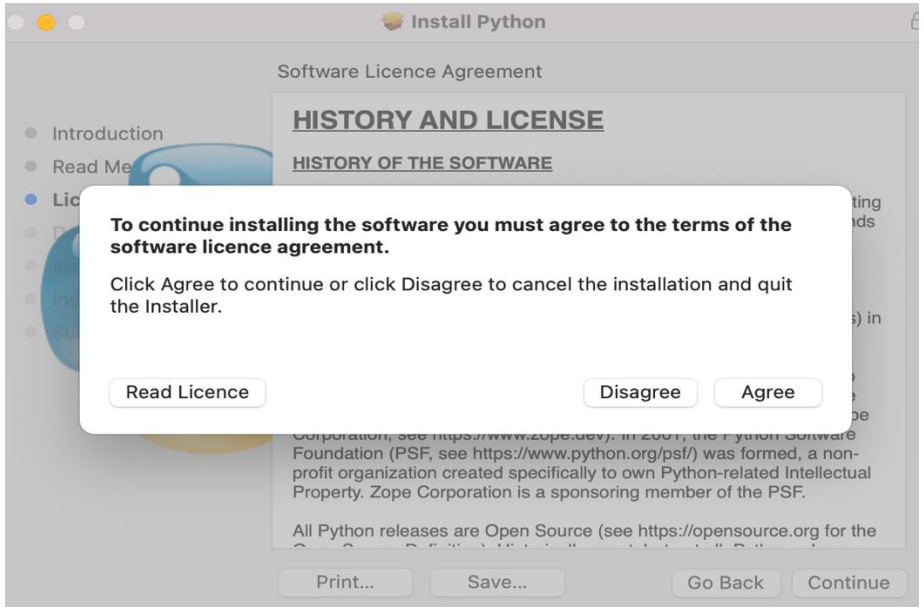
1. Double click the downloaded python-xxx.**pkg** file from the ‘Downloads’ folder
2. Click ‘**Continue**’

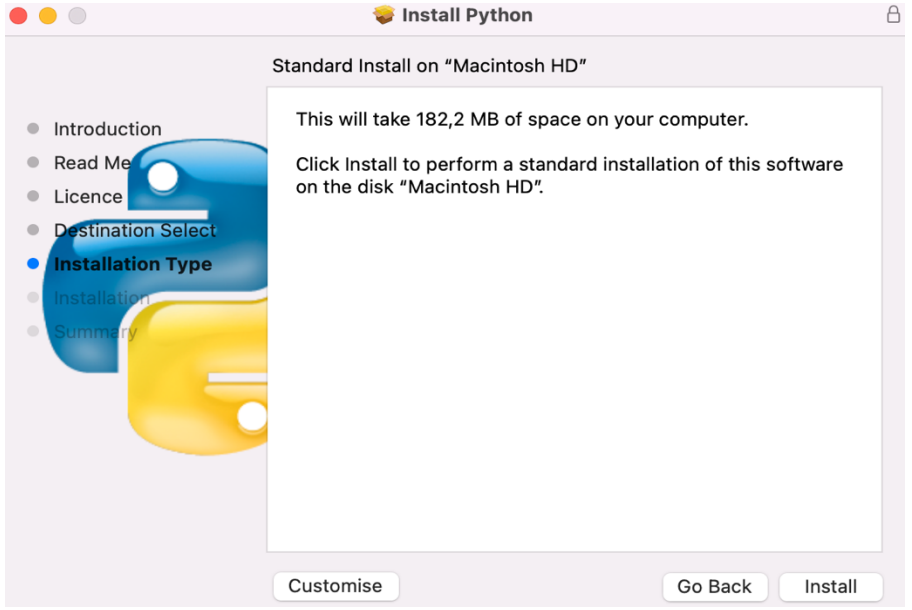




Python step-by-step guide for Mac

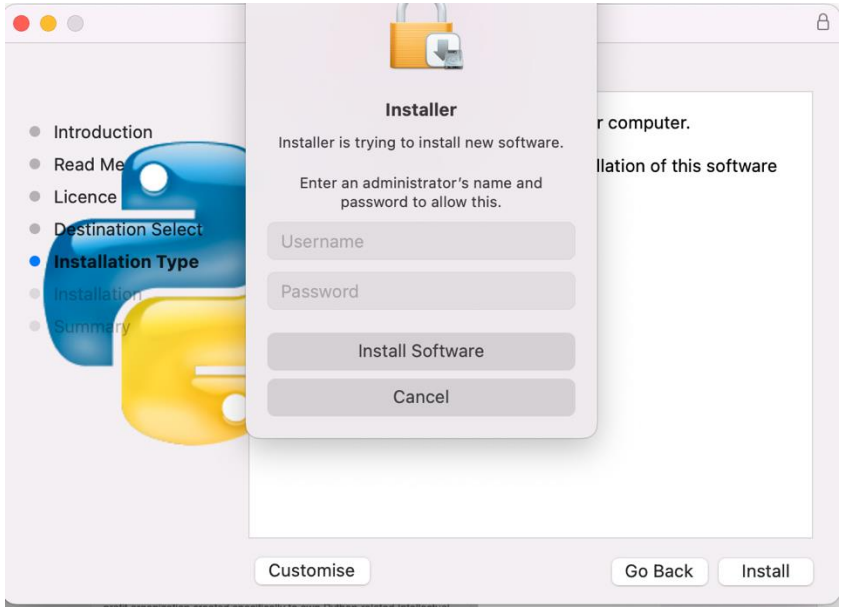
3. (Read) and Agree to the license agreement
4. Click **‘Install’**





Python step-by-step guide for Mac

5. Provide admin Username and Password, if prompted
6. Click '**Install Software**'
7. Click '**OK**' if access to folders is requested

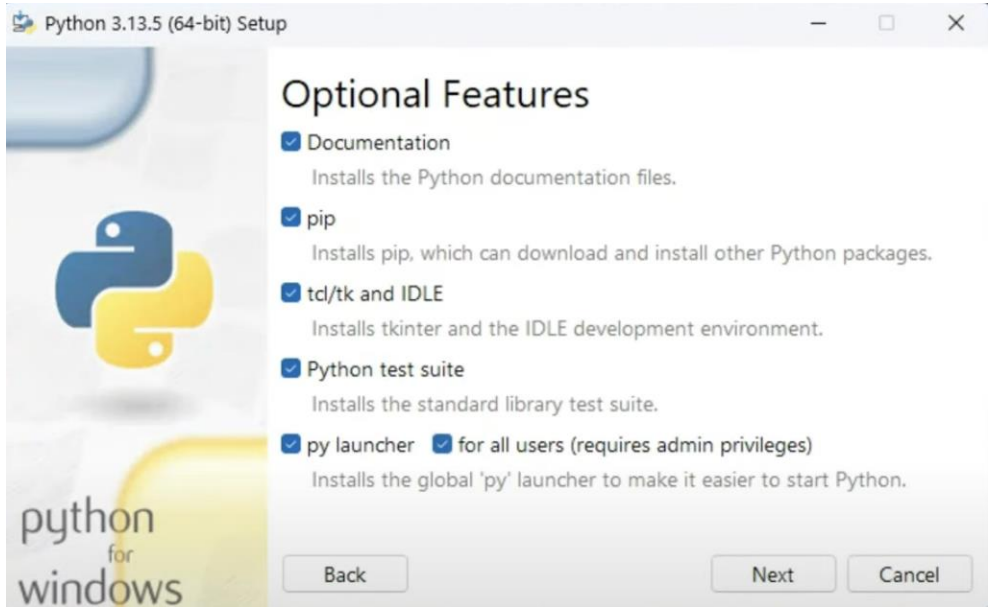




Python step-by-step guide for Windows

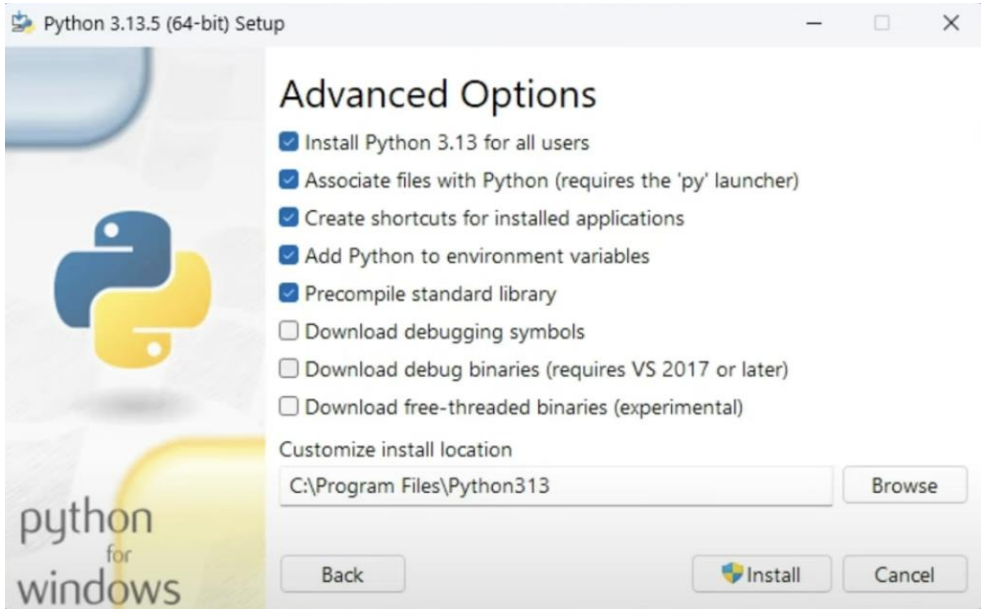
1. Double click the downloaded python-xxx64**.exe** file from the ‘Downloads’ folder
2. Ensure that the 2 checkboxes below are checked
3. Click ‘**Customize installation**’
4. Check all boxes and click ‘**Next**’

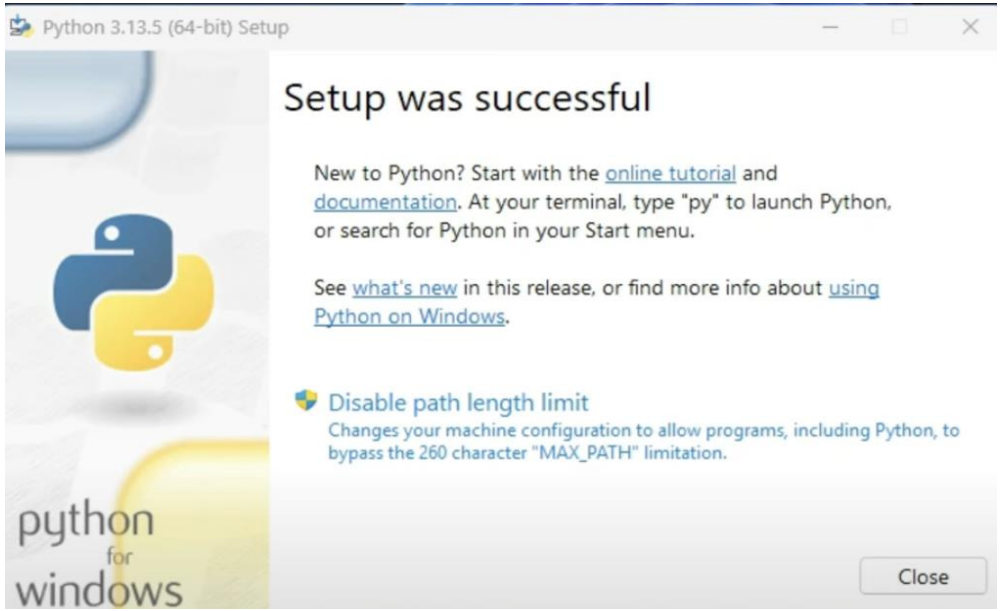




Python step-by-step guide for Windows

5. Check if the following boxes are checked and click '**Install**'
6. Click '**Close**'





Verify Python installation

Windows:

1. Press **Windows Key** or click **Start**
2. Open Command Prompt or PowerShell and type in:
   - python3 --version

   Or
   - python3 –-version

Mac:

1. Open **Launchpad**
2. Open the terminal and type in:
   - python3 --version

   Or
   - python3 –-version

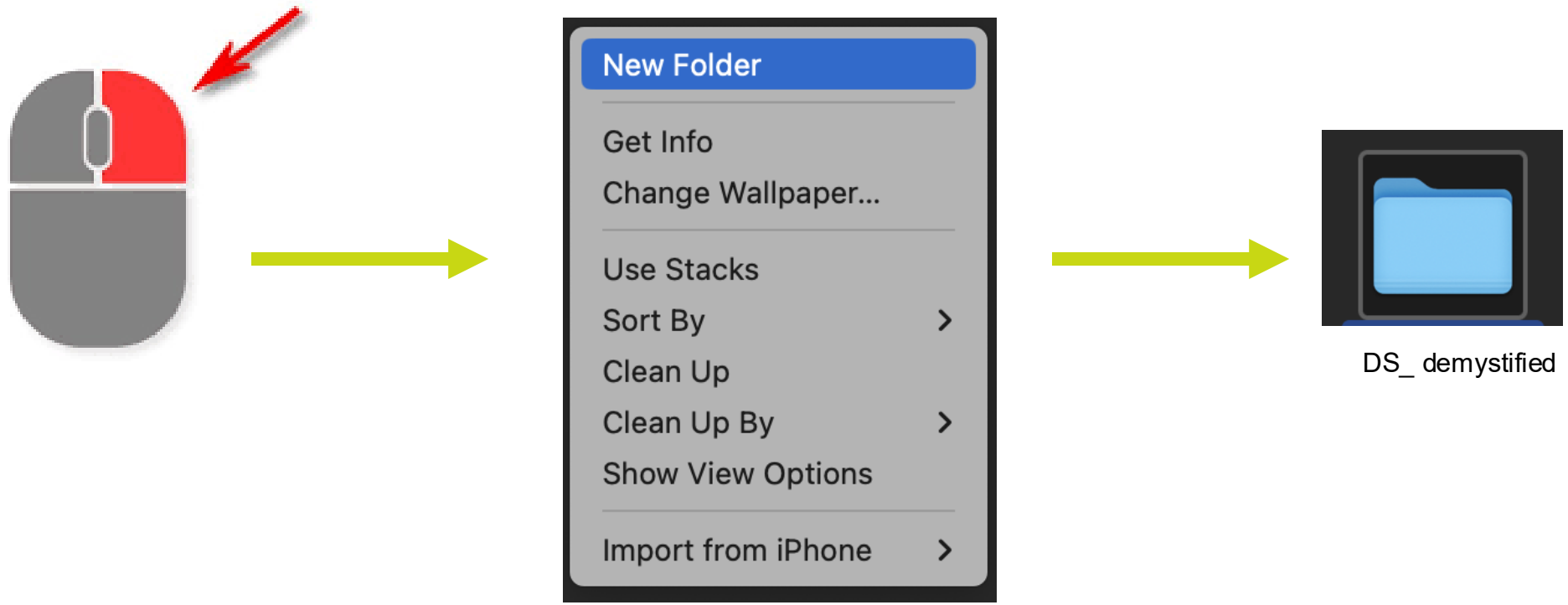
This displays the installed python version: Python 3.13.5

Virtual Environment setup



Folder setup

1. Download the provided training materials
2. Create a new folder '**DS_demystified**' on your Desktop
3. Save the course material in this folder and unzip the files



Launch and setup VS Code

1. Open the installed Visual Studio Code application with any of the following methods:

Windows:

Opt 1: Press **Windows Key** or click **Start** and select Visual Studio Code

Opt 2: Double-click the **VS Code icon** on your Desktop or Taskbar

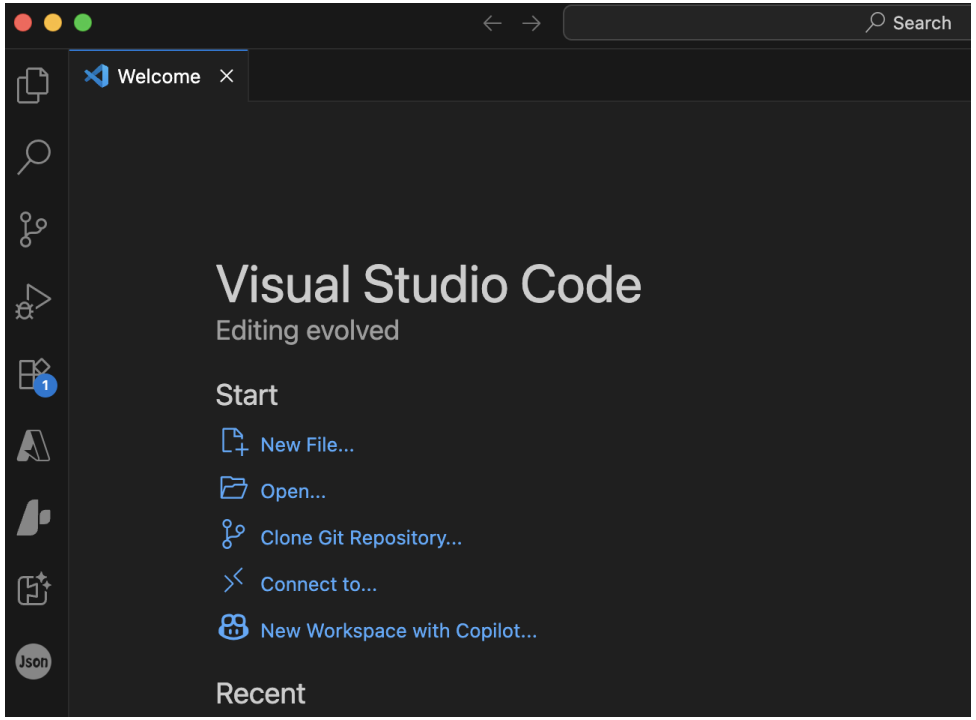
Mac:

Opt 1: Click VS Code from the Launchpad

Opt 2: Open Finder → Applications → Double-click Visual Studio Code

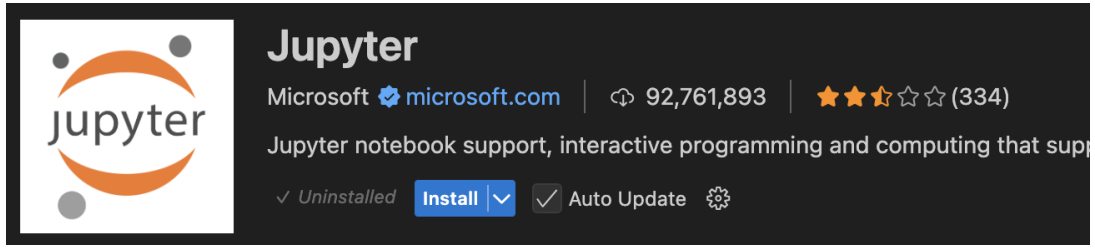
Launch and setup VS Code

2. In the '**Welcome**' screen, open the folder containing the course content with the '**Open**' option



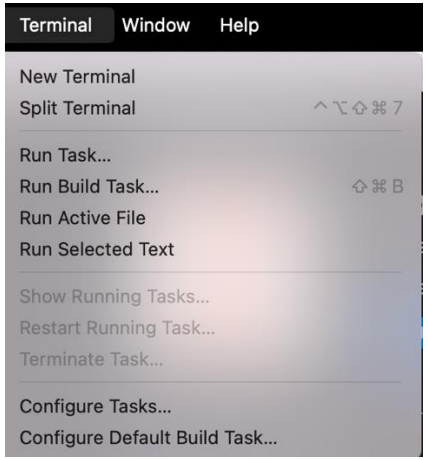
Add Jupyter extension to VS Code

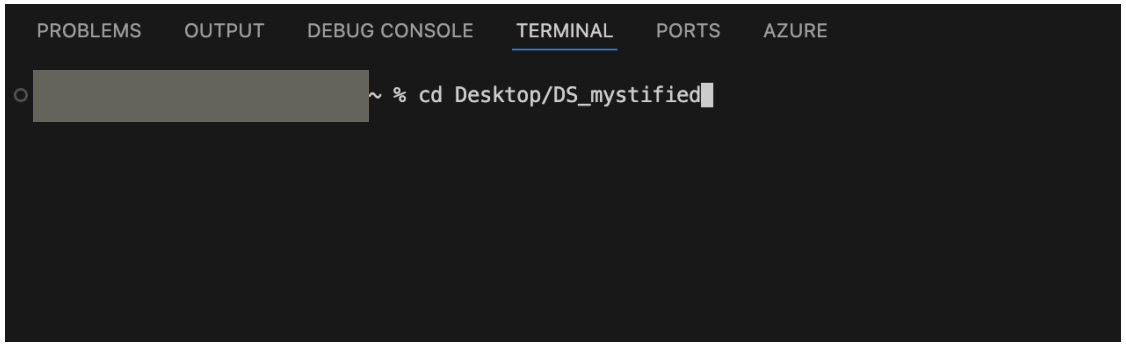
3. On the left pane of VS Code window, click on '**Extensions**'
4. Add the **Jupyter** Extension which looks like this -



Create python virtual environment

1. Click '**Terminal**' on the top menu bar of VS Code. This opens a terminal at the bottom of the VS Code window
2. In the terminal, change the working directory to DS_demystified using the '**cd**' command
   - cd Desktop/DS_ demystified



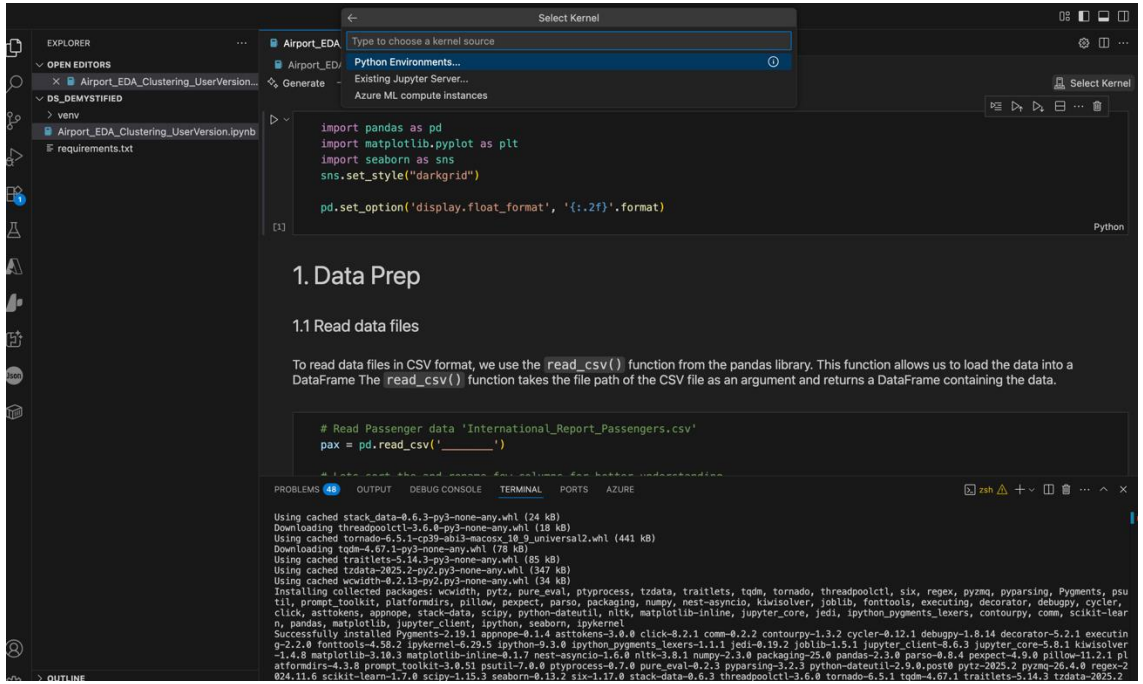


Create python virtual environment

3. **Create** a new virtual environment in the terminal
   - python3 -m venv venv
4. **Activate** the created virtual environment in the terminal
   - Windows: venv\Scripts\activate
   - Mac: source venv/bin/activate
5. **Install** the required libraries from the requirements.txt file in the terminal
   - pip install -r requirements.txt

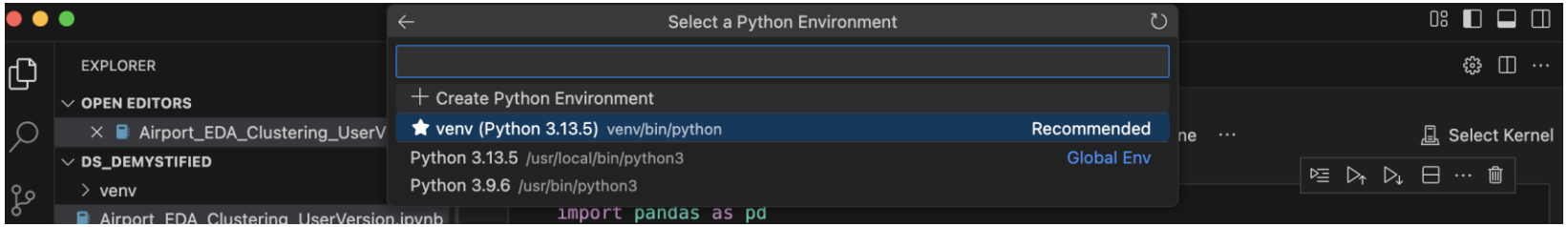
Run the Jupyter Notebook

1. Click the notebook name on the left panel to open it in the VS Code editor
2. Click **'Select Kernel'** on the top right and then **'Python Environments'** from the drop down

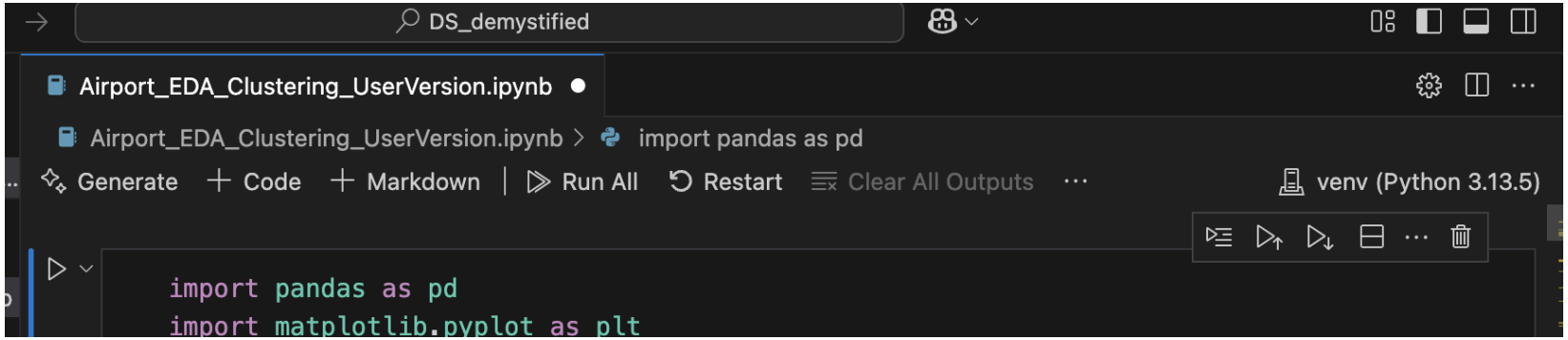


Run the Jupyter Notebook

3. Select venv (Python 3.13.5) from the dropdown

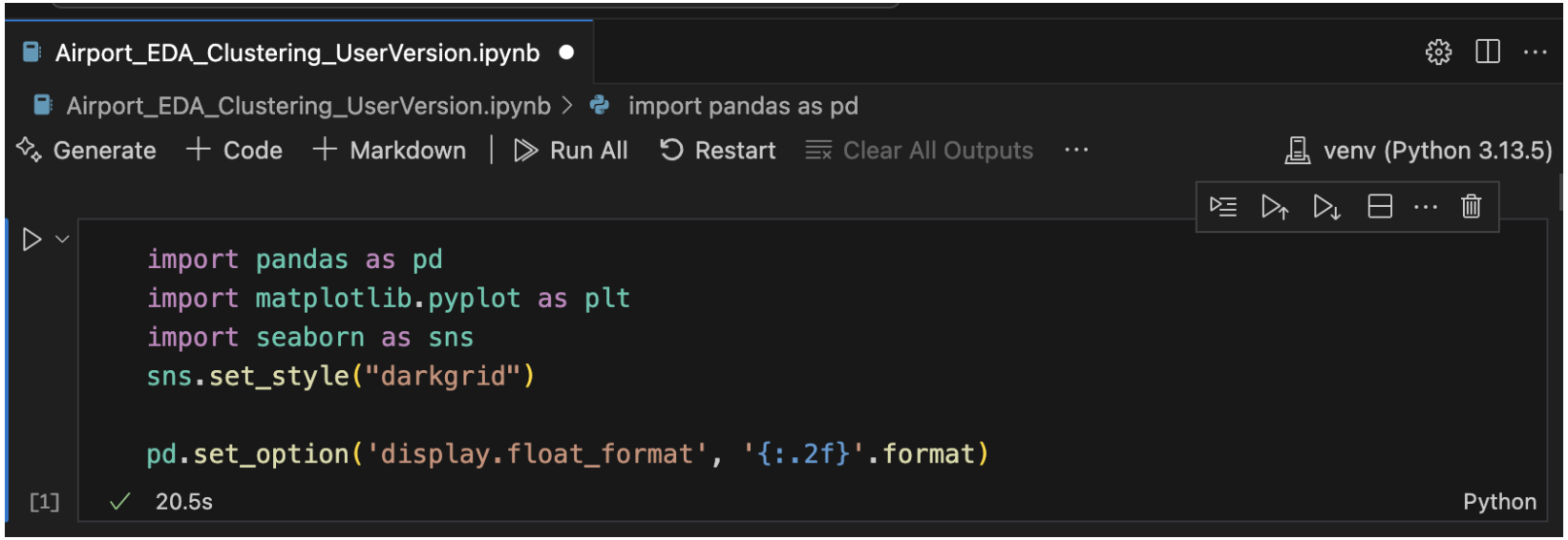


4. The selected **environment** is now reflected on the **top right corner**



Run the Jupyter Notebook

5. Click the ▶ button at the top left of each cell to run the notebook cell
6. Ensure that the cell with the imports runs without any errors



Congratulations!
You are all set for the workshop!

